Modeling Demand Curve for Non-Traditional Patterns

M.Sc. Phase-I Project (IE-685)

by

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Abstract

In economic theory, the relationship between price and quantity demanded is classically captured by the law of demand, which posits an inverse relationship: higher prices generally lead to lower demand, resulting in a negatively sloped demand curve. However, this traditional model does not fully account for complex consumer behaviors and market dynamics that shape demand in modern economies. This project investigates alternative approaches to modeling demand, particularly for products where price changes do not yield typical demand responses. We explore a generalized non-linear regression model that includes weighted variables representing price and non-price factors, such as brand strength and distribution influence, to provide a more accurate depiction of market behavior.

Our approach employs constrained regression to ensure adherence to economically meaningful principles while permitting flexibility in demand patterns. Specifically, we apply constraints to enforce a negative slope on primary demand drivers where appropriate, thus aligning with traditional economic theory, while allowing deviations in cases of unique consumer preferences or atypical elasticity. By incorporating interactions between price and brand characteristics and considering potential non-linear relationships for primary price variables, our method improves predictive accuracy and provides a robust tool for analyzing markets with intricate pricing dynamics. The findings of this study demonstrate significant enhancements in demand prediction accuracy across various product categories, underscoring the practical value of our approach in economic analysis and demand forecasting for markets exhibiting complex consumer behavior.

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Chapter 1

Introduction

The relationship between price and quantity has long been a foundational concept in economics, forming the basis of the law of demand, which asserts an inverse relationship between price and consumer demand. This principle plays a critical role in shaping pricing strategies, managing production levels, and optimizing supply chain logistics. Traditional economic models rely on the classical downward-sloping demand curve, as illustrated in Figure 1.1, where price decreases as quantity increases.

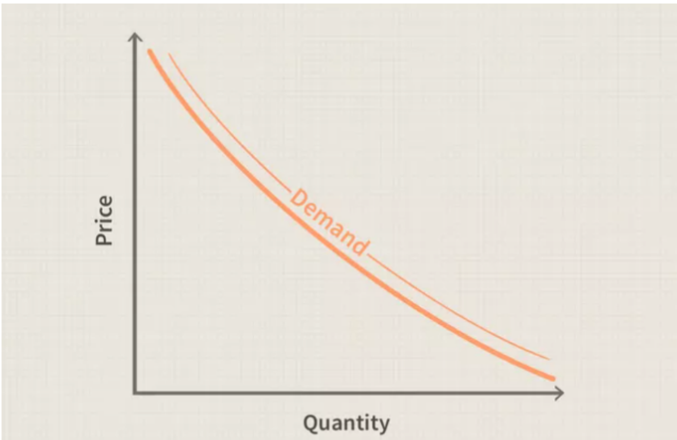


Figure 1.1: Classical demand curve showing the inverse relationship between price and quantity.

Despite its theoretical robustness, real-world markets often deviate from these assumptions. Modern consumer behavior, influenced by psychological and behavioral factors, challenges the universality of this model. Products associated with luxury or exclusivity, for example, may defy the classical demand curve, with demand remaining stable or increasing as prices rise due to perceived quality or brand prestige. This variability highlights the necessity of flexible economic models that account for diverse market conditions and consumer behaviors.

This study addresses the limitations of traditional linear demand models by proposing a non-linear regression framework. The model integrates weighted variables for price and product attributes, aligning with economic theory while capturing real-world deviations in demand behavior.

1.1 Problem Motivation

he primary motivation for this study stems from the growing disconnect between classical economic models and the realities of contemporary markets. While the classical demand curve provides a robust theoretical framework, its inability to account for behavioral and psychological factors limits its applicability in many modern contexts. Businesses today operate in an environment where consumer preferences are shaped by branding, peer influence, and perceived value rather than just price alone. As a result, traditional approaches often fail to accurately predict demand, leading to suboptimal pricing and inventory decisions.

This research aims to address these limitations by developing a non-linear regression model that integrates weighted variables for price factors and product attributes. The proposed framework not only aligns with core economic principles, such as the inverse relationship between price and demand, but also incorporates real-world deviations driven by behavioral and market-specific factors. By bridging the gap between theory and practice, this study seeks to provide a more effective tool for demand forecasting and strategic decision-making.

1.2 Key Challenges

The analysis of demand in modern markets involves several key challenges:

Consumer Behavior Variability: Modern consumers are influenced by factors such as brand perception, social influence, and loyalty, which often result in non-linear and unpredictable demand patterns. Complexity of Market Dynamics: Competitive and differentiated markets often exhibit demand behaviors that are inconsistent with classical models, requiring a more adaptive analytical framework. Incorporating Behavioral Insights: Traditional economic models largely ignore psychological and social factors, which are increasingly recognized as critical determinants of consumer behavior. To address these challenges, the study introduces a flexible non-linear regression approach that captures both price sensitivity and behavioral influences. This model incorporates constraints that ensure alignment with economic theory while allowing for deviations reflective of real-world complexities

1.3 History of Demand Curve

The concept of the demand curve, representing the relationship between price and quantity demanded, has been a cornerstone of economic theory since its formalization by Alfred Marshall in the late 19th century. Marshall's graphical representation of the inverse relationship between price and quantity helped establish the law of demand, which states that lower prices lead to higher quantities demanded, all else being equal. This foundational model has guided businesses and policymakers in decisions related to pricing, production, and resource allocation.

Throughout the 20th century, the demand curve became a vital tool for economic analysis. Businesses used it to predict consumer responses to pricing changes and design strategies to maximize revenue, while policymakers employed it to evaluate the effects of taxation, subsidies, and price controls. Despite its utility, the classical demand curve assumes a simplified world where consumer behavior is entirely rational and unaffected by external influences.

As markets grew more complex, economists began recognizing deviations from classical demand predictions. For example, luxury goods often exhibit a positive relationship between price and demand, as higher prices signal exclusivity and quality. Similarly, factors like brand loyalty, social influence, and consumer psychology challenge the notion of a uniform, downward-sloping curve. These complexities necessitated more adaptable models capable of capturing diverse market behaviors.

The mid-20th century brought significant advances with the rise of behavioral economics, led by scholars like Herbert Simon and Daniel Kahneman. Their work highlighted the role of psychological and social factors in consumer decision-making, moving beyond the purely rational assumptions of classical theory. These insights paved the way for incorporating non-linear and multi-variable approaches to demand analysis.

More recently, advancements in computational tools and data analytics have further revolutionized demand curve modeling. With access to large datasets and sophisticated techniques like non-linear regression and machine learning, analysts can now account for factors such as income, preferences, and competitor actions. These innovations allow for more precise demand predictions and enable businesses to develop highly targeted pricing strategies.

Today, the demand curve remains an essential tool for understanding consumer behavior and forecasting market trends. By integrating classical principles with modern computational methods, demand analysis has evolved into a dynamic framework for decision-making in complex markets. This study builds on this evolution, using non-linear regression techniques to enhance demand forecasting and address the challenges of contemporary markets.

Chapter 2

Literature Survey

The relationship between price and quantity, consumer surplus, price elasticities, and regression models has been extensively studied, offering valuable insights into the dynamics of price-demand interactions and consumer behavior. Choi [1] provides a theoretical exploration of consumer and producer surplus, emphasizing the impact of price variations on consumer satisfaction and producer profit margins. The study highlights how positive demand can enhance consumer surplus, while negative supply conditions often erode producer surplus. These findings are particularly relevant to our analysis, as our modeling approach seeks to incorporate the welfare implications of such price-demand relationships. This perspective aligns with the overarching goal of our research to capture demand sensitivity and its economic implications effectively.

Expanding on the concept of elasticity, the research by Hirth, Khanna, and Ruhnau [2] examines the short-term price elasticity of electricity demand in Germany. Their findings reveal significant adjustments in consumer behavior in response to immediate price changes, underscoring the importance of elasticity in understanding market dynamics. Elasticity, as defined by the percentage change in quantity demanded resulting from a one-percent change in price, plays a central role in determining how consumers respond to price shifts. This work forms the empirical basis for embedding elasticity considerations into our model, enabling us to estimate the degree of responsiveness to price fluctuations and adapt our framework to markets characterized by rapid shifts in demand.

Dooley [3], in a historical critique of Marshall's consumer surplus theory, examines alternative models for evaluating consumer satisfaction in the context of price changes. His analysis raises critical questions about accurately measuring consumer surplus and its relevance to economic welfare. By addressing the non-linear relationships between price and quantity, Dooley's research has shaped the design of our model, which integrates such complexities to better reflect real-world consumer behavior. Non-linear relationships, particularly those influenced by elasticity, are crucial for understanding how consumer surplus fluctuates with demand sensitivity and price adjustments.

Further empirical evidence is provided by Rosenthal-Kay, Traina, and Tran [4], who conducted a large-scale analysis of seven million demand elasticities across various sectors. Their research highlights the variability of demand responsiveness based on factors such as product type, market structure, and time frame. The study's findings emphasize the necessity of incorporating flexible terms in price-related features to model consumer behavior accurately. This insight has guided the incorporation of polynomial terms in our framework, allowing us to closely mimic real-world, non-linear consumer responses to price variations. By addressing sectoral and temporal heterogeneity, this approach ensures that the model reflects the diverse nature of price-demand interactions across different industries.

The literature also underscores the utility of regression models in analyzing consumer behavior, particularly in contexts where non-linear relationships dominate. Regression models provide the flexibility required to interpret complex interactions between price, competition, and demand. Building on this understanding, our research employs polynomial regression to model the subtle, non-linear effects of price on demand. This choice enables the framework to capture intricate consumer behavior patterns, ensuring that the analysis remains robust and aligned with the empirical evidence provided by prior studies.

Collectively, the reviewed literature provides a comprehensive understanding of price-demand dynamics, elasticity, and consumer surplus, offering a rich foundation for model development. The insights derived from these studies have informed our approach, motivating the inclusion of polynomial terms and constraint optimization to address real-world complexities. By synthesizing theoretical perspectives with empirical evidence, our framework integrates the nuanced effects of price and competition on consumer demand while adhering to well-established economic principles.

Chapter 3

Methodology

3.1 Generalized Weighted Metrics Calculation

Weighted metrics are commonly used to account for the relative importance or contribution of different products, brands, or categories. These calculations ensure that metrics like prices or volumes reflect their actual impact in proportion to their significance. Below, we define the generalized approach for weighted calculations.

3.1.1 Weighted Average Metric

- **Definition:** The weighted average of a metric is calculated by assigning weights proportional to another variable, such as volume or revenue. This ensures the metric reflects its real-world impact.
- **Formula:**

$$\text{Weighted Metric} = \frac{\sum_{i=1}^{n} W_i M_i}{\sum_{i=1}^{n} W_i}$$

where:

  - M_i: The metric being averaged (e.g., price, sales growth).
  - W_i: The weight associated with M_i (e.g., volume, revenue).
  - n: The total number of observations (e.g., products, brands).

- **Interpretation:** The numerator aggregates the contribution of each observation weighted by its significance, while the denominator normalizes the total weight.

3.1.2 Weighted Price Calculation

- **Purpose:** To compute the average price across products, accounting for their relative sales volume.
- **Formula:**

$$P_{\text{weighted}} = \frac{\sum_{i=1}^{n} P_i V_i}{\sum_{i=1}^{n} V_i}$$

  where:

  - P_i: The price of product i.
  - V_i: The volume sold of product i.
  - n: The total number of products.

- **Interpretation:** This metric captures the average price while considering that products with higher sales volumes contribute more to the overall average.

3.1.3 Weighted Volume Share

- **Purpose:** To determine the proportion of total volume contributed by a specific product, brand, or category.
- **Formula:**

$$VS_i = \frac{V_i}{\sum_{j=1}^{n} V_j} \times 100$$

  where:

  - VS_i: The volume share of product i.
  - V_i: The volume sold of product i.
  - n: The total number of products or brands.

- **Interpretation:** This metric represents the percentage contribution of a product's volume to the total market volume.

3.1.4 Weighted Metrics for Index Calculations

- **Category Price Index (CPI):** A normalized price metric that accounts for the sales volume of each product.

$$CPI_t = \frac{\sum_{i=1}^{n} P_{i,t} V_{i,t}}{\sum_{i=1}^{n} V_{i,t}}$$

  where:

- $P_{i,t}$: Price of product i at time t.
- $V_{i,t}$: Volume of product i at time t.

- **Weighted Price Index:** To measure relative price movement across periods.

$$\text{Price Index} = \frac{\sum_{i=1}^{n} P_{i,t} V_{i,t}}{\sum_{i=1}^{n} P_{i,0} V_{i,0}}$$

where:

- $P_{i,t}$: Price of product i at time t.
- $V_{i,t}$: Volume of product i at time t.
- $P_{i,0}, V_{i,0}$: Base price and volume for product i at the baseline period.

3.1.5 Metric Creation: Analyzing Consumer Behavior and Price Sensitivity

To understand how consumers behave over time with respect to price—specifically, whether they are gravitating towards costlier or cheaper products—we design a metric that removes the mix effect of volume. This approach is detailed below.

The first step involves preparing the data by focusing on key variables: *Price*, *Volume*, and *Brands/Variants*. The data is then pivoted such that brands/variants become columns, while the price and volume values are aggregated accordingly. This pivot table serves as the basis for further calculations.

Next, we compute the weighted price for each brand or variant. For a specific brand i, the weighted price is calculated as:

$$P_{weighted,i} = \frac{\sum_{i=1}^{n} P_i V_i}{\sum_{i=1}^{n} V_i},$$

where P_i is the price of brand i, V_i is its corresponding volume, and n is the total number of brands/variants in the group. This step provides insight into how the volume shifts between different brands at their respective prices.

After calculating the weighted price for each brand, we move to an aggregate horizontal analysis. This involves summing the weighted prices across all brands in a row to calculate the overall category-level price for a specific time period. The formula for this aggregated price is:

$$P_{category} = \frac{\sum_{i=1}^{n} (P_i \cdot V_i)}{\sum_{i=1}^{n} V_i}.$$

This aggregated value represents the weighted average price across all brands, normalizing the impact of individual brand volumes.

Once we have the aggregated category-level price, we create the **Up-Down Price Index** by plotting $P_{category}$ over time. The resulting curve captures continuous price movement, indicating whether consumers are shifting towards higher or lower-priced products within the category.

To normalize the category-level trends, we compute the **Category Price Index (CPI)**. This is achieved by dividing the Up-Down Price Index by its baseline, which is the average of the Up-Down Index over the time period. The CPI is calculated as:

$$\text{CPI} = \frac{P_{category}}{\overline{P_{category}}},$$

where $\overline{P_{category}}$ is the mean of $P_{category}$ over time.

The CPI provides3.1 a clear indication of consumer trends. If the CPI is greater than 1, it indicates that the category is shifting towards more expensive products; if it is less than 1, the category is moving towards cheaper alternatives.

By following this approach, we obtain metrics that offer actionable insights into consumer price sensitivity and behavior over time, enabling informed decisions about pricing and market strategy.

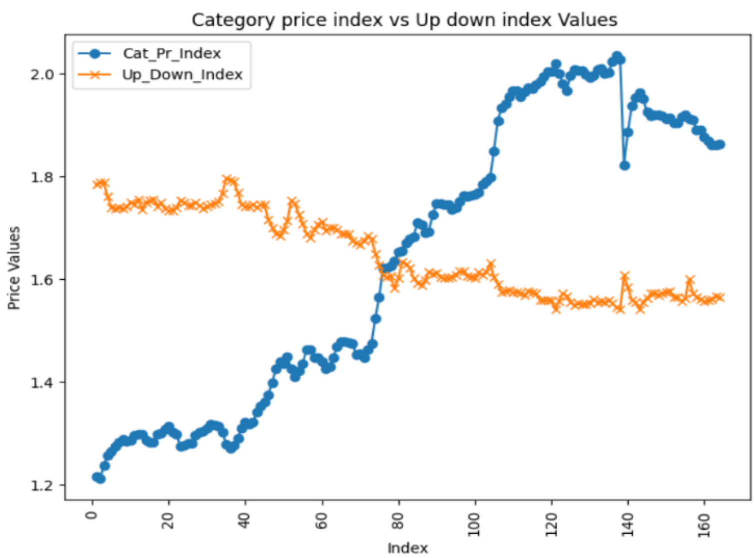


Figure 3.1: Up Down Index

3.1.6 Modeling Features

To capture the behaviors of markets and consumers in different areas, we developed and used the following features in our analysis. Each feature combines theoretical insights with mathematical formulations to clarify their role and implementation.

1. **Price (PPL)**

   - **Description:** Price per Liter (PPL) is a key factor in understanding the performance of brands, packs, or variants in a category.
   - **Purpose:** According to the demand curve principle, increasing price generally decreases volume, revealing price sensitivity.
   - **Method:** To capture non-linear relationships between price and volume, we model the relationship using a cubic function:

     $$V = \beta_0 + \beta_1 P + \beta_2 P^2 + \beta_3 P^3 + \epsilon$$

     where V represents volume, P is the price, $\beta_0, \beta_1, \beta_2, \beta_3$ are coefficients, and ϵ is the error term. This approach helps uncover complex consumer responses to pricing.

2. **Competitive Pricing**

   - **Cannibalization Price:** Measures how the price of one product affects the volume of another within the same brand or pack. The relationship is modeled as:

     $$V_i = \gamma_0 + \gamma_1 P_c + \epsilon$$

     where P_c is the price of the competing product, and V_i is the volume of the analyzed product.
   - **Direct Competition Price:** Quantifies the impact of prices from other brands offering similar products in the same category:

     $$V_i = \delta_0 + \delta_1 P_d + \epsilon$$

     where P_d represents the price of direct competitors.
   - **Indirect Competition Price:** Captures the effect of cross-category competition by products serving similar consumer needs:

     $$V_i = \phi_0 + \phi_1 P_{ind} + \epsilon$$

     where P_{ind} is the price of an indirect competitor.

3. **Seasonality, Trend, and Category Volume**

   - **Seasonality:** Reflects demand fluctuations across specific periods (e.g., weeks or months). Seasonal volume is calculated as:

     $$S_t = \sum_{i=1}^{n} V_{i,t}$$

     where S_t is the seasonal volume at time t, and n is the number of products.

   - **Trend:** Captures long-term directional changes in demand. The trend (T) is derived as a function of time t:

     $$T = f(t)$$

   - **Category Volume:** Represents the total demand in a category over a specific time:

     $$CV_t = \sum_{i=1}^{n} V_{i,t}$$

     where CV_t is the category volume at time t.

4. **Distribution**

   - **Cannibalization Distribution:** Shows the percentage of overlap in sales among products of the same brand:

     $$D_c = \frac{V_i}{\sum_{j=1}^{m} V_j} \times 100$$

     where V_i is the volume of a product, and m is the number of products in the brand.

   - **Direct Competition Distribution:** Reflects competitive brand availability:

     $$D_d = \frac{\text{Availability of competing products}}{\text{Total Availability}} \times 100$$

   - **Indirect Competition Distribution:** Assesses market competition across categories:

     $$D_{ind} = \frac{\text{Overlap in consumer reach}}{\text{Total market reach}} \times 100$$

5. **Up-Down Index and Category Price Index**

- **Up-Down Index:** Tracks shifts in consumer behavior towards higher or lower price products. It is calculated as:

$$UDI = \frac{\sum_{i=1}^{n} P_i V_i}{\sum_{i=1}^{n} V_i}$$

where P_i is the price, and V_i is the volume of the i-th product.

- **Category Price Index:** Monitors average price levels in a category over time:

$$CPI_t = \frac{\sum_{i=1}^{n} P_{i,t} V_{i,t}}{\sum_{i=1}^{n} V_{i,t}}$$

By comparing UDI and CPI, we assess whether price shifts align with overall category trends.

6. **Brand Value and Interaction Terms**

- **Brand Value:** Encoded as binary variables to identify brand-specific effects:

$$B_{i,j} = \begin{cases} 1 & \text{if product } i \text{ belongs to brand } j, \\ 0 & \text{otherwise.} \end{cases}$$

- **Interaction Terms:** Capture brand-specific price sensitivity through interactions between price and brand:

$$V = \alpha_0 + \alpha_1 (P \times B) + \epsilon$$

where α_1 represents the unique impact of price for each brand.

Each feature is designed to provide a comprehensive understanding of market dynamics, enabling better prediction of consumer behavior and sales volumes. By combining theoretical insights and mathematical formulations, we capture complex interactions among price, competition, seasonality, and brand dynamics.

3.2 Mathematical Model

In general, we can represent the regression model as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \tag{3.1}$$

where y is the dependent variable (sales volume) and x_i represents the various independent variables or features, with their respective coefficients β_i.

3.2.1 Approach 1: Linear Model for PPG-Level Analysis by Brand

In our case, the model includes price sensitivity, competition, seasonality, trend, and brand interaction terms. Given that the model operates on the PPG level, we consider three PPG categories (Small, Medium, and Large) and four brands (A, B, C, D). The equation for the sales volume Q can be written as:

$$
\begin{aligned}
Q = \beta_0 &+ \beta_1 \,\text{Price (PPL)} + \beta_2 \,\text{Price (Cannibalization)} + \beta_3 \,\text{Price (Direct Competition)} + \beta_4 \,\text{Price (I} \\
&+ \beta_5 \,\text{Distribution (Distance)} + \beta_6 \,\text{Distribution (Cannibalization)} + \beta_7 \,\text{Distribution (Direct Com} \\
&+ \beta_9 \,\text{Seasonality} + \beta_{10} \,\text{Trend} + \beta_{11} \,\text{Up-Down Index} + \beta_{12} \,\text{Category Price Index} \\
&+ \beta_{13} \,\text{Category Volume} + \beta_{14} \,\text{Brand A (One-Hot Encoded)} + \beta_{15} \,\text{Brand B (One-Hot Encoded)} \\
&+ \beta_{16} \,\text{Brand C (One-Hot Encoded)} + \beta_{17} \,\text{Brand D (One-Hot Encoded)} \\
&+ \beta_{18} \,\text{Price (PPL)} \times \text{Brand A} + \beta_{19} \,\text{Price (PPL)} \times \text{Brand B} \\
&+ \beta_{20} \,\text{Price (PPL)} \times \text{Brand C} + \beta_{21} \,\text{Price (PPL)} \times \text{Brand D} + \epsilon
\end{aligned}
$$

where:

- Q: The dependent variable, representing the predicted sales volume.
- $\beta_0, \beta_1, \ldots, \beta_{21}$: Coefficients representing the strength of each feature's impact on sales volume.
- ϵ: The error term, capturing any variation not explained by the model.

Each coefficient β_i reflects the strength of each feature's impact on the sales volume Q. The error term ϵ accounts for any variations not explained by the model.

*

Constraint Optimization for Regression Model

To solve this linear regression problem with specific constraints on the coefficients, we applied a **constrained optimization** approach. Here's a summary of the constraints we used for each type of feature, particularly focusing on the pricing variables to capture the required relationships:

- **Price (PPL)**: The coefficient for *Price (PPL)*, represented by β_1, was constrained to be **negative**. This reflects the intuitive expectation that, as the price per liter increases, the sales volume Q would generally decrease.
- **Price (Cannibalization), Price (Direct Competition), and Price (Indirect Competition)**: The coefficients for these terms—β_2, β_3, and β_4—were constrained to be **positive**. This captures the assumption that competitive or cross-category effects should have a positive impact on the sales of the main brand, as increases in competitors' prices make our product relatively more attractive.

To implement these constraints, we used a **constrained optimization algorithm** (such as Sequential Least Squares Quadratic Programming, SLSQP), which allows us to set boundaries on specific coefficients during the model training. This approach ensured that the model adhered to these business-driven assumptions, providing interpretable and meaningful coefficients that align with economic theory and practical insights.

So, the optimization problem can be formulated as follows:

$$\text{Minimize: } \sum (Q - \hat{Q})^2$$

Subject to:

$$\begin{aligned} &\beta_1 \leq 0 && \text{(for Price (PPL) to be negative)} \\ &\beta_2, \beta_3, \beta_4 \geq 0 && \text{(for other price terms to be positive)} \end{aligned}$$

This constraint-based approach allowed us to effectively capture the expected market behavior while fitting the model to the data.

3.2.2 Approach 2: Polynomial Regression with Degree 3 for Price (PPL)

In the second approach, we introduce **non-linearity for the Price (PPL) feature** by using a polynomial of degree 3. This adjustment acknowledges that the relationship between the price per liter (PPL) and sales volume may not be strictly linear; instead, it may exhibit diminishing or accelerating effects at different price levels.

By applying a polynomial transformation only to the *Price (PPL)* feature, we allow for greater flexibility in modeling the influence of price changes while maintaining a linear relationship for other price-related and non-price features. The remaining features, including ”Price (Cannibalization),” ”Price (Direct Competition),” and ”Price (Indirect Competition),” retain their linear effects to reflect simpler, more direct influences on sales.

The modified regression model with polynomial terms for *Price (PPL)* is expressed as:

$$
\begin{aligned}
Q = {} & \beta_0 + \beta_1 \,\text{Price (PPL)} + \beta_2 \,\text{Price (PPL)}^2 + \beta_3 \,\text{Price (PPL)}^3 \\
& + \beta_4 \,\text{Price (Cannibalization)} + \beta_5 \,\text{Price (Direct Competition)} + \beta_6 \,\text{Price (Indirect Competition)} \\
& + \beta_7 \,\text{Distribution (Distance)} + \beta_8 \,\text{Distribution (Cannibalization)} + \beta_9 \,\text{Distribution (Direct Com} \\
& + \beta_{10} \,\text{Distribution (Indirect Competition)} + \beta_{11} \,\text{Seasonality} + \beta_{12} \,\text{Trend} \\
& + \beta_{13} \,\text{Up-Down Index} + \beta_{14} \,\text{Category Price Index} + \beta_{15} \,\text{Category Volume} \\
& + \beta_{16} \,\text{Brand A (One-Hot Encoded)} + \beta_{17} \,\text{Brand B (One-Hot Encoded)} \\
& + \beta_{18} \,\text{Brand C (One-Hot Encoded)} + \beta_{19} \,\text{Brand D (One-Hot Encoded)} \\
& + \beta_{20} \,\text{Price (PPL)} \times \text{Brand A} + \beta_{21} \,\text{Price (PPL)} \times \text{Brand B} \\
& + \beta_{22} \,\text{Price (PPL)} \times \text{Brand C} + \beta_{23} \,\text{Price (PPL)} \times \text{Brand D} + \epsilon
\end{aligned}
$$

Constraints for Polynomial Model

The constraints on the coefficients remain aligned with the business requirements set in the first approach, with additional considerations for the non-linear Price (PPL) terms:

- **Price (PPL)**: The coefficients β_1, β_2, and β_3 for *Price (PPL)* are constrained to maintain an overall **negative influence on sales volume** as Price (PPL) increases. This captures the assumption that higher product prices generally reduce demand.

- **Price (Cannibalization), Price (Direct Competition), and Price (Indirect Competition)**: These coefficients remain **positive**, indicating that higher prices of competing products could positively influence the sales of our product due to reduced competition.

By incorporating non-linearity only in the primary price feature, *Price (PPL)*, we achieve a model that balances flexibility in capturing complex price dynamics with interpretability and control over the other price-related factors.

3.2.3 Approach 3: Custom Ridge Regression with PPL Transformation

In this approach, we implemented a Custom Ridge Regression model designed to handle brand-level sales volume predictions while accounting for price sensitivity, seasonality, and brand-specific interactions. Key transformations applied to the primary price variable (`PPL`) aim to better capture non-linear relationships across different brands and channels.

Model Improvements Over Previous Approaches

The following improvements distinguish this approach from the previous two:

- **Transformation of Price Variable (PPL)**: We apply different transformations to the `PPL` variable to enhance the model's flexibility and capture non-linear relationships. Specifically, we tested reciprocal, logarithmic, exponential, and linear transformations, selecting the one with the lowest MAPE.
- **Interaction Terms for Brand and PPL**: By introducing interaction terms between transformed `PPL` values and encoded brand categories, the model adapts to varying price sensitivities across brands.
- **Constraints and Regularization**: We utilize L2 regularization (Ridge) along with constraints on feature weights, improving model stability and preventing overfitting.

3.2.4 Workflow

The workflow for this approach is detailed below:

Algorithm 1 Custom Ridge Regression with Brand Interaction Terms

Input:

Dataset: `df_ch_ppg`

Target variable: `Volume`

Initialize:

Transformation functions: {`reciprocal, exp_neg, linear`}

Predefined L2 penalty values

Procedure:

for `transform_function` ∈ {`reciprocal, exp_neg, linear`} **do**

Transform the `PPL` feature using the `transform_function`

One-hot encode the `Brand` column and create interaction terms with `PPL_transformed`

Standardize all features

Optimization:

for `l2_penalty` in predefined range **do**

Define the Ridge-regularized cost function

Optimize parameters using the SLSQP algorithm with weight constraints

Calculate training and test MAPE (Mean Absolute Percentage Error)

Update Best Parameters:

if training MAPE improves **then**

Update the best parameters

Record the minimum MAPE

Output:

Best parameters

Overall MAPE

Brand-wise MAPE

3.3 Demand Curve Transformations

The demand curve equations are derived using different transformations applied to the price per liter (PPL). These transformations allow the relationship between price and demand to exhibit various behaviors, including linear, reciprocal, and exponential relationships. The final demand curve incorporates the effects of other variables (X_i) through their mean values and coefficients.

3.3.1 Reciprocal Transformation

The reciprocal transformation assumes an inverse relationship between price and demand. The transformation applied to PPL is:

$$PPL_{\text{transformed}} = \frac{1}{PPL}$$

The corresponding demand curve equation is:

$$Q = \beta_0 + \sum_{i=2}^{23} \beta_i X_i + \beta_1 \cdot \frac{1}{PPL}$$

This indicates that as PPL increases, the transformed value decreases, leading to a steep decline in demand, particularly at low price points.

3.3.2 Negative Exponential Transformation

The negative exponential transformation models demand as exponentially decreasing with price. The transformation applied is:

$$PPL_{\text{transformed}} = e^{-PPL}$$

The demand curve equation becomes:

$$Q = \beta_0 + \sum_{i=2}^{23} \beta_i X_i + \beta_1 \cdot e^{-PPL}$$

This transformation highlights rapid changes in demand for moderate price adjustments, capturing an exponential decay behavior.

3.3.3 Linear Transformation (No Transformation)

For the linear case, the PPL is used without any transformation. The transformation applied is:

$$PPL_{\text{transformed}} = PPL$$

The demand curve equation is:

$$Q = \beta_0 + \sum_{i=2}^{23} \beta_i X_i + \beta_1 \cdot PPL$$

This assumes a direct and consistent relationship between price and demand, with a constant rate of change in Q for variations in PPL.

For all transformations, the effects of other variables (X_i) such as Competition Variables, Distribution Variables, Seasonality etc. are treated as constants. These variables are incorporated into the demand curve equation through their mean values and respective coefficients β_i. This ensures that the final demand curve focuses on the relationship between price and quantity demanded while accounting for the average contribution of other factors.

3.4 Price Elasticity

From the demand curves, the price elasticity of demand (ε) was calculated for each transformation to understand how quantity demanded (Q) changes with price (PPL). The elasticity formula is expressed as:

$$\varepsilon = \frac{\partial Q}{\partial PPL} \cdot \frac{PPL}{Q}$$

This equation captures the percentage change in quantity demanded resulting from a one-percent change in price. The calculation has several implications:

1. **Elastic vs. Inelastic Demand:**

    - If $|\varepsilon| > 1$, demand is considered elastic, meaning consumers are highly responsive to price changes. In such cases, small changes in price lead to larger proportional changes in demand.
    - If $|\varepsilon| < 1$, demand is inelastic, indicating that consumers are less responsive to price changes. Here, price increases can often lead to revenue gains despite a reduction in quantity sold.
    - If $|\varepsilon| = 1$, demand is unit elastic, where price changes result in proportional changes in quantity demanded, leaving revenue unchanged.

2. **Transformation Impact on Elasticity:** Each transformation applied to PPL changes the functional form of the demand curve and influences how $\frac{\partial Q}{\partial PPL}$ behaves:

    - **Linear Transformation:** Elasticity remains constant along the demand curve. Changes in price directly translate to proportional changes in quantity.

- **Negative Exponential Transformation:** Elasticity varies with price. Lower prices tend to exhibit higher elasticity as demand is more sensitive at these levels, while higher prices lead to diminishing responsiveness.
- **Reciprocal Transformation:** Elasticity decreases as price increases. This behavior indicates that price increases at higher levels have a smaller relative impact on demand, which may suggest opportunities for premium pricing.

3. **Practical Relevance:**
   - Elasticity provides a robust framework for assessing revenue opportunities. For products with elastic demand, price reductions can stimulate demand and potentially boost overall revenue. Conversely, for inelastic demand, increasing prices might maximize profitability with limited impact on sales volume.
   - Understanding elasticity across the transformations enables businesses to evaluate not only the current pricing strategy but also the potential trade-offs between price, demand, and revenue for different PPG categories.

4. **Dynamic Nature of Elasticity:**
   - Elasticity is not a static property. It varies depending on the price level and market conditions, as illustrated by the different transformations. This underscores the need for continuous market monitoring and adjustment of pricing strategies to align with consumer behavior.

Chapter 4

Experiments and Results

4.1 Data Set

In this project, we utilized real-world data from multiple sectors. To maintain confidentiality, details of specific firms and their product are omitted. The dataset primarily includes:

- **Numerical Columns:** The key numerical variables are *Price* and *Quantity*.
- **Categorical Columns:** Various categorical fields such as *PPG* (Promoted Product Group), *PackType*, *Brand*, *Channel*, *Market*, and sometimes *Distribution*.

Since categorical data cannot be directly applied to models, we performed feature engineering to create new variables that could better capture the underlying relationships. For our analysis, we primarily focused on *PPG/PackType* groups and conducted a detailed examination across brands within these groupings.

4.2 Experiments

Evaluation

We evaluated the constrained model using the Mean Absolute Percentage Error (MAPE) metric for both training and test datasets. This method allowed us to maintain the interpretability of the demand curve while ensuring acceptable prediction accuracy. Further experiments explored the impact of different functions and regularization techniques on model performance.

Considerations

The graphs presented in this study correspond to a single combination of channel, `PPG`, and brands, as the data contains a total of 23 possible combinations, as shown in the table. While the results generally showed minor variation, there was some fluctuation due to the train-test split. These variations are expected and highlight the sensitivity of the model to the data partitioning process and due to privacy concern we are not map the original labels of channel, brand, ppg to numbers as in table but in below table to understand brands (A,B,C,D) as brand Private Label kept remain same to understand that data used for experiment is real . For demand predictions, we employed the Rectified Linear Unit (ReLU) function, which helped address non-linearity in the data. The ReLU function, defined as:

$$\text{ReLU}(x) = \max(0, x)$$

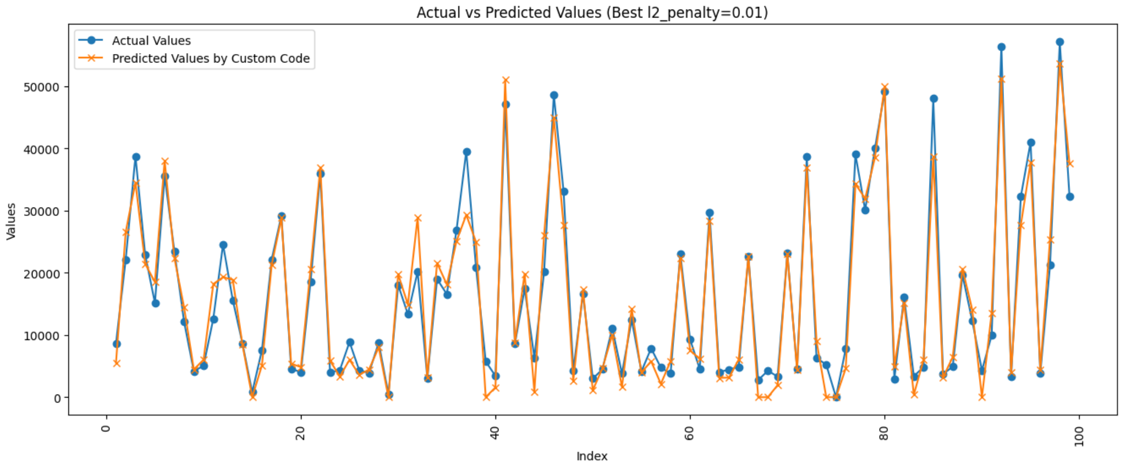
was applied to ensure that only non-negative predictions were made, aligning with the demand curve's typical behavior.

4.2.1 Linear Ridge Regression

Initially, we employed simple linear regression to predict `Volume` based on `PPG` and other pricing features. The results showed that for some `PPG/Pack` combinations, the model produced reasonable predictions. However, a critical issue arose as the weights (β coefficients) for `PPG` were not always negative, violating the basic assumption of a negatively sloped demand curve. To resolve this, we implemented a constrained regression approach, ensuring that the coefficient for `PPG` (`PPL`) remained negative, reflecting the expected inverse relationship between price and demand. Other price-related coefficients were constrained to be positive, in line with practical expectations. The optimization process minimized MAPE while adhering to these constraints.

Linear Regression: Experimental Results

The results in Figure 4.1b summarize the MAPE values for training and testing across different channels, PPG, and brands. Additionally, Figure 4.1a shows the fit plot for testing data , allowing for a visual assessment of the model's performance on both datasets.



(a) Fit Plot (Test Data) for Ridge Regression

Channel	PPG	Product Brands	Train_MAPE	Test_MAPE
Channel 1	PPG 1	['Brand A', 'Brand B', 'Other Brands']	14.82	17.84
Channel 1	PPG 2	['Brand A', 'Brand B', 'Private Label Std']	12.63	14.76
Channel 1	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	14.17	13.7
Channel 1	PPG 4	['Brand C', 'Brand B', 'Brand D', 'Other Brands']	11.84	12.55
Channel 1	PPG 5	['Brand B']	8.83	10.98
Channel 1	PPG 6	['Brand B']	8.43	10.3
Channel 2	PPG 5	['Brand A', 'Brand B']	16.0	11.06
Channel 2	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	15.92	17.06
Channel 2	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Other Brands']	11.92	10.77
Channel 2	PPG 2	['Brand A', 'Brand B']	18.03	21.17
Channel 2	PPG 1	['Brand B']	7.38	8.78
Channel 3	PPG 5	['Brand A', 'Brand B']	10.2	11.66
Channel 3	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	13.3	12.24
Channel 3	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	11.99	11.04
Channel 3	PPG 2	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	15.48	16.04
Channel 3	PPG 6	['Brand C', 'Brand B']	11.28	11.47
Channel 3	PPG 1	['Brand B', 'Other Brands']	14.64	12.12
Channel 4	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	12.71	13.87
Channel 4	PPG 2	['Brand A', 'Brand B', 'Brand D']	10.47	10.44
Channel 4	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Brand E', 'Other Brands']	16.57	15.59
Channel 4	PPG 1	['Brand B']	12.42	10.22
Channel 4	PPG 5	['Brand B']	12.57	17.09
Channel 4	PPG 6	['Brand B']	16.16	14.22

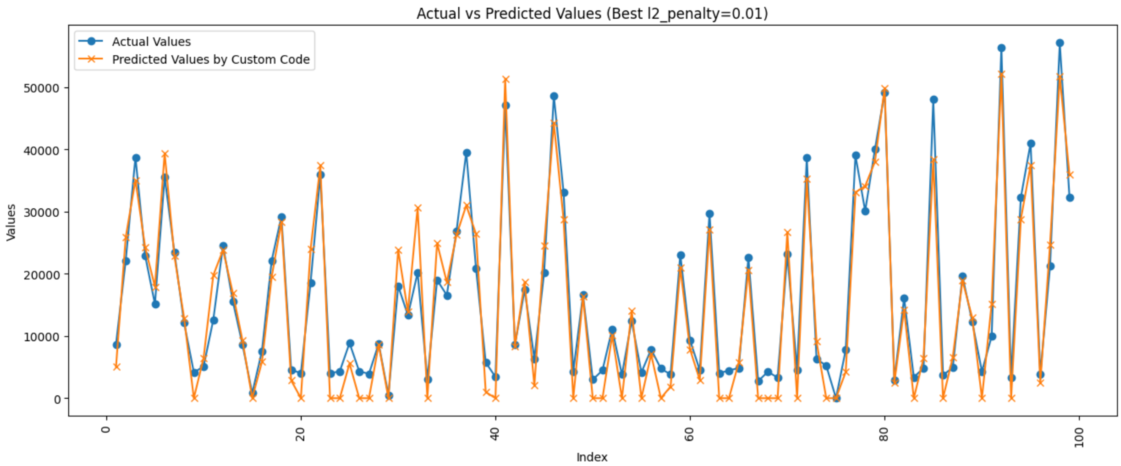
(b) Results Table for Different Channels, PPGs, and Brands

4.2.2 Custom Ridge Regression with transformation

Reciprocal Transformation: Experimental Results

For the reciprocal transformation, no negative constraints were necessary on the PPL feature, unlike the linear transformation. This improved model performance, as reflected in the significantly lower MAPE for both the training and test sets, compared to the linear case. However, it should be noted that for certain combinations of PPG and brand levels, the reciprocal transformation did not perform as well, especially where the relationship between price and volume is more complex.

The corresponding fit plot and MAPE results for the reciprocal transformation are shown below:

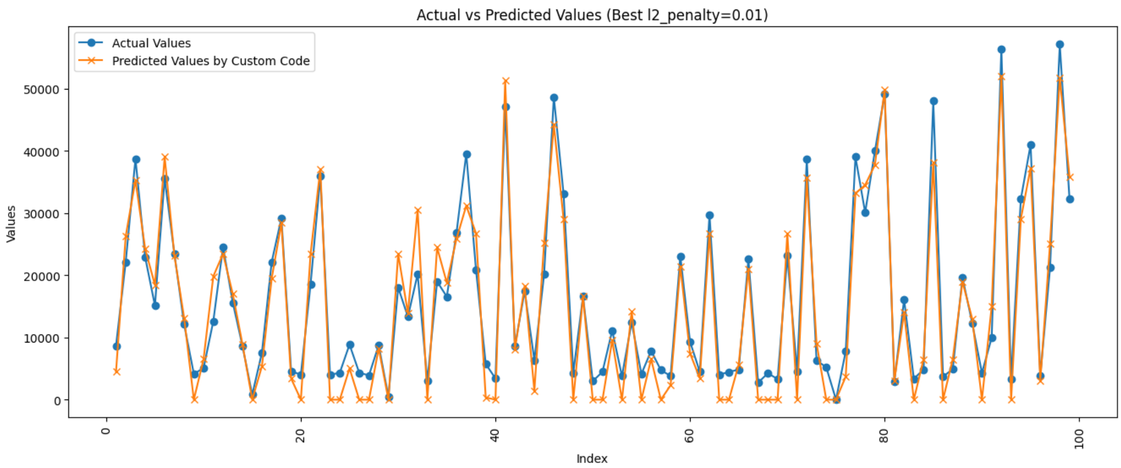


(a) Fit Plot (Test Data) for Reciprocal Ridge Regression

Channel	PPG	Product Brands	Train_MAPE	Test_MAPE
Channel 1	PPG 1	['Brand A', 'Brand B', 'Other Brands']	15.94	18.65
Channel 1	PPG 2	['Brand A', 'Brand B', 'Brand C']	16.02	19.27
Channel 1	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D']	14.98	15.71
Channel 1	PPG 4	['Brand B', 'Brand C', 'Brand D']	14.45	14.71
Channel 1	PPG 5	['Brand B']	8.85	10.89
Channel 1	PPG 6	['Brand B']	8.17	10.21
Channel 2	PPG 5	['Brand A', 'Brand B']	14.51	10.18
Channel 2	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	18.31	18.85
Channel 2	PPG 2	['Brand A', 'Brand B']	11.23	10.02
Channel 2	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Other Brands']	19.5	22.67
Channel 2	PPG 6	['Brand B']	7.4	8.82
Channel 3	PPG 5	['Brand A', 'Brand B']	11.75	12.42
Channel 3	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	14.29	13.1
Channel 3	PPG 2	['Brand A', 'Brand C', 'Brand B', 'Brand D']	12.8	12.12
Channel 3	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	16.46	16.43
Channel 3	PPG 6	['Brand C', 'Brand B']	11.93	11.05
Channel 3	PPG 1	['Brand B', 'Other Brands']	14.92	11.96
Channel 4	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Brand D']	12.88	13.38
Channel 4	PPG 2	['Brand A', 'Brand B', 'Brand C']	11.06	11.04
Channel 4	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Brand D', 'Other Brands']	18.66	18.73
Channel 4	PPG 1	['Brand B']	12.14	9.91
Channel 4	PPG 5	['Brand B']	13.12	16.91
Channel 4	PPG 6	['Brand B']	15.63	13.29

(b) Results Table for Reciprocal Transformation - MAPE across Channels, PPGs, and Brands

Negative Exponential Transformation: Experimental Results



(a) Fit Plot (Test Data) for Negative Exponential Ridge Regression

Channel	PPG	Product Brands	Train_MAPE	Test_MAPE
Channel 1	PPG 1	['Brand A', 'Brand B', 'Other Brands']	15.53	17.34
Channel 1	PPG 2	['Brand A', 'Brand B', 'Private Label Std']	15.92	19.44
Channel 1	PPG 3	['Brand A', 'Brand C', 'Brand B', 'Private Label Std']	15.2	15.84
Channel 1	PPG 4	['Brand C', 'Brand B', 'Private Label Std']	14.98	14.85
Channel 1	PPG 5	['Brand B']	9.94	11.57
Channel 1	PPG 6	['Brand B']	7.79	9.98
Channel 2	PPG 1	['Brand A', 'Brand B']	14.86	10.35
Channel 2	PPG 2	['Brand A', 'Brand C', 'Brand B']	18.54	19.98
Channel 2	PPG 3	['Brand A', 'Brand B']	11.67	10.32
Channel 2	PPG 4	['Brand A', 'Brand C', 'Brand B']	19.84	22.17
Channel 2	PPG 5	['Brand B']	7.47	8.88
Channel 3	PPG 1	['Brand A', 'Brand B']	11.9	12.53
Channel 3	PPG 2	['Brand A', 'Brand C', 'Brand B']	14.63	13.44
Channel 3	PPG 3	['Brand A', 'Brand B', 'Private Label Std']	12.98	12.1
Channel 3	PPG 4	['Brand A', 'Brand C', 'Brand B', 'Private Label Std']	17.79	17.95
Channel 3	PPG 5	['Brand C', 'Brand B']	11.78	11.0
Channel 3	PPG 6	['Brand B']	14.92	11.95
Channel 4	PPG 1	['Brand A', 'Brand B']	12.98	13.52
Channel 4	PPG 2	['Brand A', 'Brand B']	10.91	10.79
Channel 4	PPG 3	['Brand A', 'Brand C', 'Brand B']	19.59	19.08
Channel 4	PPG 4	['Brand A', 'Brand B']	12.14	9.98
Channel 4	PPG 5	['Brand A', 'Brand B']	13.32	16.95
Channel 4	PPG 6	['Brand B']	15.37	13.17

(b) Results Table for Negative Exponential Transformation - MAPE across Channels, PPGs, and Brands

In comparison to the reciprocal transformation, the negative exponential approach proved more effective for certain PPG and brand combinations. This transformation naturally captured the downward-sloping demand curve, which is particularly useful in situations where there is a clear inverse relationship between price and volume. The negative exponential transformation did not require any additional constraints on the coefficients, and it led to better predictive performance in cases where the reciprocal transformation struggled to provide accurate results.

For these cases, the negative exponential transformation resulted in a more robust model with lower MAPE values across several key channels and PPG levels, as shown in the fit plot and MAPE table below.

4.3 Results

4.3.1 Demand Curve Visualization

To better understand consumer behavior, three distinct demand curves were generated based on transformations applied to the price per liter (PPL). These transformations—Linear, Negative Exponential, and Reciprocal—offer insights into how the quantity demanded (Q) varies with changes in price under different assumptions. Each curve highlights a unique aspect of the price-demand relationship and provides a foundation for analyzing price elasticity and pricing strategies.

Each curve not only captures a different behavioral aspect of the market but also lays the groundwork for calculating price elasticity, which is crucial for making informed pricing decisions.

Price Elasticity Analysis: Linear Transformation

Understanding Elasticity for the Linear Transformation: The price elasticity of demand (ε) is a critical metric that quantifies the responsiveness of quantity demanded (Q) to changes in price (PPL). For the Linear Transformation, the demand curve takes the form:

$$Q = a - b \cdot PPL$$

where:

- a: Intercept, representing the theoretical maximum demand when the price is zero.
- b: Slope, indicating the rate of decline in demand with respect to price.

The elasticity formula is:

$$\varepsilon = \frac{\partial Q}{\partial PPL} \cdot \frac{PPL}{Q} = -b \cdot \frac{PPL}{a - b \cdot PPL}.$$

Results from Linear Transformation:

- **Constant Rate of Decline:** The linear demand curve assumes a uniform sensitivity across all price levels. As price increases, the demand decreases at a constant rate, leading to predictable changes in revenue and demand.
- **Elasticity Variation with Price:** While the slope of the demand curve (b) remains constant, the elasticity (ε) changes depending on the price level. At lower prices, the demand is less elastic (consumers are less sensitive to price changes), while at higher prices, demand becomes more elastic.
  - At lower price points, businesses may experience smaller relative changes in demand when increasing prices, making slight price hikes feasible for revenue maximization.
  - At higher price points, any increase in price leads to a relatively larger drop in demand, requiring careful evaluation of pricing strategies.

Behavior of the Linear Demand Curve: The demand curve generated under the Linear Transformation is depicted in **Figure 4.4**. The curve represents a straight line, demonstrating a consistent relationship between price and quantity demanded. This transformation is often employed when consumer behavior does not show significant variation in sensitivity across price ranges.

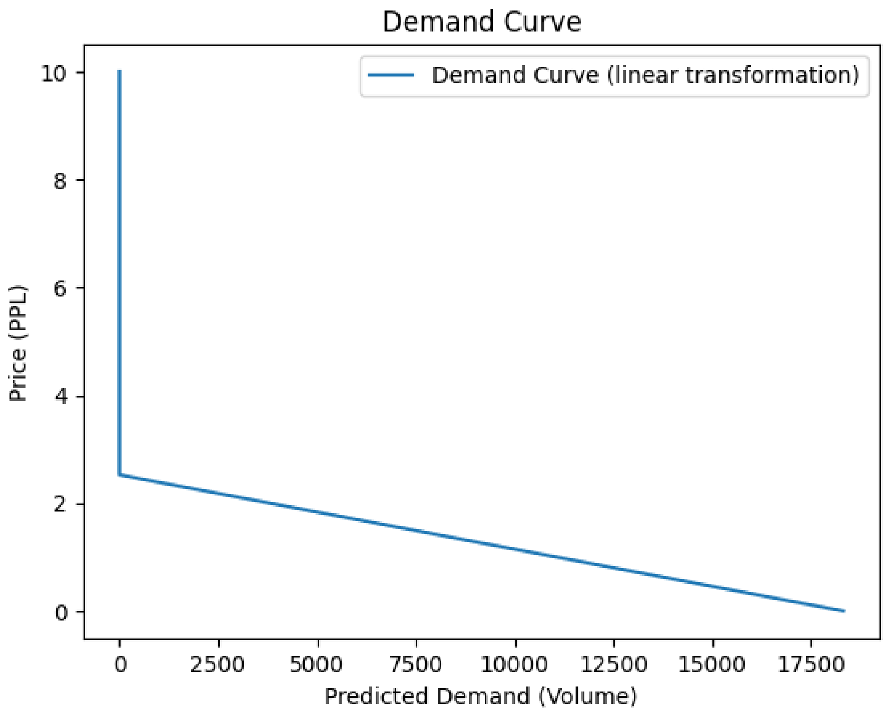


Figure 4.4: Demand Curve - Linear Transformation

Negative Exponential Transformation:

The demand curve shown in **Figure 4.5** represents the Negative Exponential Transformation, where we observe a rapid initial decline in demand as the price increases, with small changes in price leading to significant decreases in quantity demanded. This pattern highlights a market where price sensitivity is high at lower price points. As the price rises, however, the effect of price increases on demand starts to diminish, causing the curve to flatten out. This behavior suggests that consumers are highly responsive to price changes when the price is relatively low, but as prices increase, their sensitivity decreases.

The Negative Exponential Transformation is particularly useful for understanding markets where products exhibit high price sensitivity at lower price levels but become less responsive to price changes at higher levels. In such markets, companies must be cautious when increasing prices at lower price points, as small price hikes can lead to a significant loss in demand. However, once prices reach a certain threshold, the impact of further price increases on demand is less pronounced, which opens up opportunities for pricing strategies that aim for higher margins.

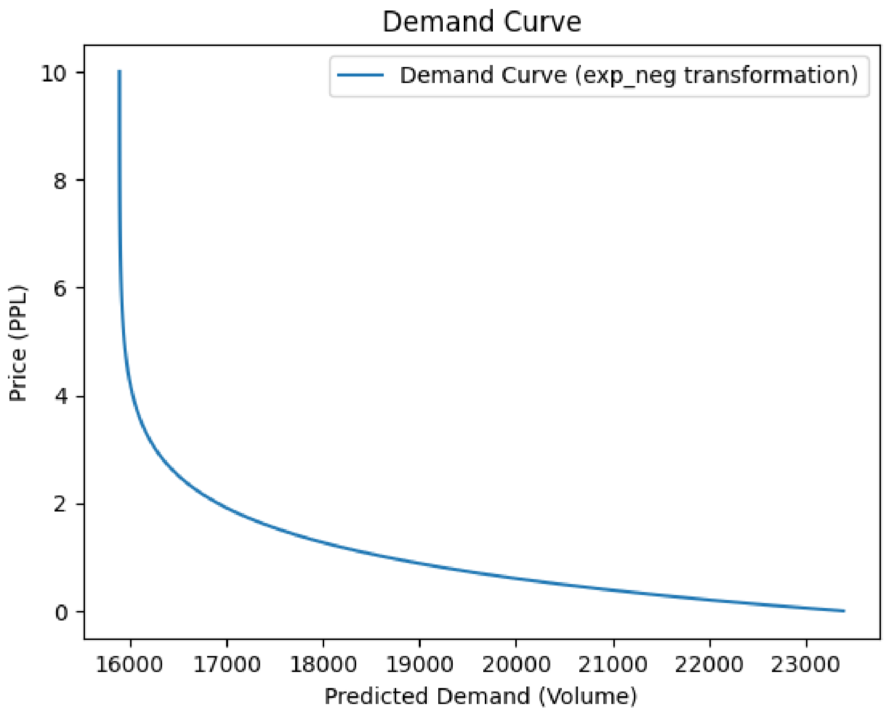


Figure 4.5: Demand Curve - Negative Exponential Transformation

As shown in **Figure 4.5**, the demand curve steeply declines at first, illustrating the sensitivity of consumers to price changes in the early stages. But as the price continues to rise, the curve flattens, indicating that further price increases have a diminishing effect on demand. This flattening of the curve reflects reduced price elasticity at higher price levels, meaning that at certain price points, the market becomes less responsive to price changes.

This characteristic of the Negative Exponential demand curve is crucial for setting pricing strategies. For products that follow this demand pattern, businesses can consider implementing price hikes cautiously in the beginning, as consumers may be highly sensitive to price changes. However, once the price reaches a certain point, there may be room for price increases without a significant loss in demand, making it possible to maximize revenue by focusing on premium pricing strategies at higher price ranges.

Reciprocal Transformation:

The demand curve depicted in **Figure 4.6** represents the Reciprocal Transformation, where demand experiences steep changes at lower price levels but gradually flattens out as the price increases. This behavior suggests that consumers are highly sensitive to price changes when prices are low, but as prices rise, their responsiveness diminishes. Essentially, the curve indicates that, at higher price points, demand stabilizes, and further price increases have less of an impact on consumer behavior.

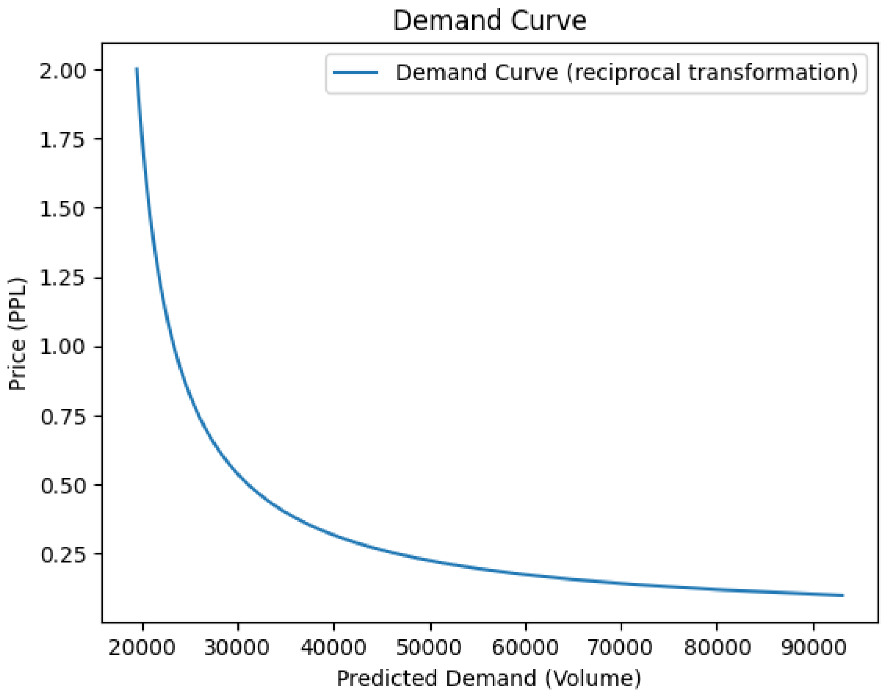


Figure 4.6: Demand Curve - Reciprocal Transformation

In terms of price elasticity, the reciprocal transformation typically results in a demand curve that reflects a diminishing rate of change in demand relative to price. Initially, small price hikes cause significant decreases in demand, but as the price increases, the elasticity tends to decrease, showing a reduction in price sensitivity at higher price levels. This type of curve is typical for products where consumers may show strong sensitivity at the lower end of the price range but exhibit greater tolerance for price increases once a certain price threshold is crossed.

The Reciprocal Transformation is useful for markets where products can sustain price increases at higher levels without significantly reducing demand. This characteristic is particularly beneficial for businesses looking to implement premium pricing strategies, as they can raise prices with a reduced risk of losing customers once prices reach a certain threshold. The demand curve flattens as prices move up, signaling that consumers are less responsive to further price hikes in these higher ranges.

4.3.2 Results of Optimal Demand Curve (Particular PPG) for Analysis

To identify the best demand curve for each Price Point Group (PPG) and brand combination in a specific channel or market, the models were evaluated based on their Train Mean Absolute Percentage Error (MAPE). The demand curve with the lowest Train MAPE was considered optimal for that combination, as it provides the most accurate representation of the relationship between price and quantity demanded.

Criteria for Selection

The following steps were used to determine the best demand curve:

1. Calculate the Train MAPE for each transformation: Linear, Negative Exponential, and Reciprocal.

2. Compare the Train MAPE values across the transformations for each PPG and brand combination in a given channel or market.

3. Select the transformation with the lowest Train MAPE as the optimal demand curve for that combination.

4.3.3 Final Interpretation

Table 4.7 summarizes the selected optimal transformations for key combinations of PPGs, brands, and channels, along with their Train MAPE values. The optimal demand curves are then used to analyze pricing strategies and revenue potential.

Channel	PPG	Unique Brands	Train MAPE	Test MAPE	Best Transformation
Channel 1	PPG 6	Brand A, Brand B, Other Brands	16.1	18.94	exp_neg
Channel 1	PPG 3	Brand A, Brand B, Brand D	12.8	15.67	exp_neg
Channel 1	PPG 4	Brand A, Brand C, Brand B, Brand E	13.92	15.16	reciprocal
Channel 1	PPG 2	Brand C, Brand B, Brand E	11.01	11.24	reciprocal
Channel 1	PPG 1	Brand B	8.83	10.94	reciprocal
Channel 1	PPG 5	Brand B	7.79	10.05	exp_neg
Channel 2	PPG 1	Brand A, Brand B	13.87	10.06	exp_neg
Channel 2	PPG 2	Brand A, Brand C, Brand B, Brand E	18.28	18.56	reciprocal
Channel 2	PPG 3	Brand A, Brand B	11.42	10.09	reciprocal
Channel 2	PPG 4	Brand A, Brand C, Brand B, Other Brands	13.08	18.39	exp_neg
Channel 2	PPG 5	Brand B	7.4	8.81	reciprocal
Channel 3	PPG 1	Brand A, Brand B	9.98	10.91	reciprocal
Channel 3	PPG 2	Brand A, Brand C, Brand B, Brand E	11.18	10.16	reciprocal
Channel 3	PPG 3	Brand A, Brand C, Brand B, Brand D	12.81	11.79	reciprocal
Channel 3	PPG 4	Brand A, Brand C, Brand B, Brand D	11.66	12.56	exp_neg
Channel 3	PPG 5	Brand C, Brand B	10.99	10.66	exp_neg
Channel 3	PPG 6	Brand B, Other Brands	14.02	12.26	exp_neg
Channel 4	PPG 2	Brand A, Brand C, Brand B, Brand E	10.82	12.1	reciprocal
Channel 4	PPG 3	Brand A, Brand B, Brand D	10.94	10.83	exp_neg
Channel 4	PPG 4	Brand A, Brand C, Brand B, Brand E	13.38	12.68	exp_neg
Channel 4	PPG 6	Brand B	12.14	9.96	exp_neg
Channel 4	PPG 1	Brand B	13.14	16.98	reciprocal
Channel 4	PPG 5	Brand B	15.37	13.24	exp_neg

Figure 4.7: Best Transformation use to plot demand Curve

The table presented in **Figure 4.7** illustrates the best transformation selected for each Promoted Price Group (PPG)/ Packtype and brand combination based on the lowest Train Mean Absolute Percentage Error (MAPE). The "Best Transformation" column of the table indicates which demand curve (linear, negative exponential, or reciprocal) provides the most accurate representation of price and quantity demanded for each PPG-brand combination.

The selection of the optimal demand curve for each combination enables targeted analysis of price sensitivity and revenue optimization. The combination of accurate demand modeling and elasticity analysis provides a comprehensive framework for strategic decision-making in pricing and product positioning across different channels and markets.

Using the selected demand curves, we can now predict future market behavior with greater confidence. The insights gained from the elasticity analysis of each transformation provide a deeper understanding of how consumers are likely to react to price changes moving forward. For instance, in PPGs where the reciprocal transformation has been selected as the best fit, we anticipate that future price increases will have a reduced impact on demand at higher price points. Conversely, in categories where the negative exponential transformation is optimal, we expect that even small price increases could lead to significant drops in demand, particularly at lower price points.

This prediction framework, supported by the chosen demand curves, allows market/business to optimize their pricing strategies based on the expected future responses of consumers to price changes. By forecasting demand elasticity under different pricing scenarios, companies can make more informed decisions about price adjustments, promotional strategies, and product positioning. Furthermore, the elasticity insights derived from the demand curves offer a basis for long-term revenue optimization and market trend forecasting.

4.3.4 Generalization of Approach

To validate the robustness of the methodology, we applied this analysis across more than five real-world datasets, encompassing various Price Groups (PPGs), pack types, and brand variants. In each case, we were able to derive an optimal demand curve that best represented the price-quantity relationship within the given data. This demonstrates that our approach is not data-specific but rather a generalized methodology capable of capturing consumer behavior across a wide range of market conditions and product categories. By identifying the most appropriate transformation for each dataset, our approach ensures a broader applicability and is adaptable to different PPGs, variants, and channels.

Chapter 5

Conclusion and Future works

5.1 Conclusion

Through our analysis, we have observed distinct patterns in the demand curves across various Price Groups (PPGs). Products with linear demand relationships show a moderate responsiveness to price changes, suggesting that price increases, when carefully managed, can lead to incremental revenue gains without significant losses in demand. In contrast, PPGs characterized by steep elasticity, particularly those identified under the negative exponential transformation, reveal a much stronger sensitivity to price increases. For these categories, price hikes lead to a disproportionate decline in demand, indicating that strategies focused on increasing sales volume rather than raising prices will be more effective. Finally, products exhibiting reciprocal demand curves, especially at higher price points, demonstrate resilience to price changes. These flatter demand curves suggest that these products can withstand price increases with minimal impact on demand, positioning them as ideal candidates for premium pricing strategies.

The elasticity results provide actionable insights into pricing strategies for each PPG category in the market. For products with a linear demand relationship, revenue increases are likely to be achievable through a careful balance between price increases and modest reductions in demand. Products exhibiting negative exponential demand curves, however, require caution, as significant price hikes could lead to substantial losses in demand, making them unsuitable for revenue optimization through price increases. In contrast, products with reciprocal demand relationships show resilience to price changes, particularly at higher price points, offering opportunities to enhance profitability without significant declines in sales.

This analysis not only supports data-driven pricing decisions but also offers a solid quantitative basis for maximizing profitability in the market. By understanding the specific elasticity dynamics of each PPG and leveraging the optimal demand curve for each, businesses can craft more effective pricing strategies that align with consumer behavior, competitive pressures, and market trends. To capitalize on these insights, companies should regularly reassess the price elasticity of their product categories, monitor changes in consumer demand, and refine their pricing models to stay ahead of market shifts. Ultimately, the ability to predict and manage price sensitivity across various categories enables companies to strategically position their products, optimize revenue, and achieve long-term profitability.

5.2 Future Work

An extension of this analysis could incorporate competitive pricing dynamics to further refine our elasticity and demand curve models. By including relative prices from competitors, we can better capture the competitive landscape and its effect on demand. The proposed transformation for price would take the form of:

$$p + \left(\frac{p}{p_{\text{comp1}}}\right) + \left(\frac{p}{p_{\text{comp2}}}\right),$$

where the price of the product is adjusted relative to the prices of competing products. This approach considers how fluctuations in competitors' pricing affect consumer demand for our product, allowing for a more nuanced understanding of elasticity in a competitive context.

Additionally, in this framework, we would maintain a common price across all competitors, treating the relative prices as constant factors. However, the demand curve will feature a unique β coefficient for each product, reflecting the specific price sensitivity and elasticity of demand based on the competitive pricing landscape. This will allow for more tailored pricing strategies that take into account not only consumer preferences and elasticity but also the influence of competition in the market.

By integrating competitive pricing into our analysis, we aim to provide businesses with a more robust tool for optimizing pricing strategies in dynamic, competitive markets. Future work could also explore variations in consumer behavior, market share shifts, and product differentiation to refine the model further.

Bibliography

[1] Choi, H. (2021). The Profitable Theory of Consumers' and Producers' Surplus: Positive Demand, Negative Supply?. Advisor, Chung-Hua Institution for Economic Research.

[2] Hirth, L., Khanna, T., Ruhnau, O. (2022). The (very) short-term price elasticity of German electricity demand. ZBW - Leibniz Information Centre for Economics, Kiel, Hamburg. Available at: `http://hdl.handle.net/10419/249570`.

[3] Dooley, P. C. (1983). Consumer's Surplus: Marshall and His Critics. *The Canadian Journal of Economics / Revue canadienne d'Economique*, 16(1), 26-38. Published by Wiley on behalf of the Canadian Economics Association. Available at: `https://www.jstor.org/stable/134973`.

[4] Rosenthal-Kay, J., Traina, J., Tran, U. (2021). Seven million demand elasticities. Leibniz Information Centre for Economics.

[5] Neelamegham, R., Chintagunta, P. (2003). A Bayesian Model to Forecast New Product Performance in Domestic and International Markets. *INSEAD, Graduate School of Business, University of Chicago*. Available at: `https://www.insead.edu`.

[6] Manchanda, P., Rossi, P. E., Chintagunta, P. K. (2004). Investigating Heterogeneity in Brand Preferences in Logit Models for Panel Data, Response Modeling with Nonrandom Marketing-Mix Variables. *Journal of Marketing Research*, 41(4), 467-4700. 10.1509/jmkr.41.4.467.4700.

[7] Hitsch, G. J., Hortaçsu, A., Lin, X. (2019). Prices and promotions in U.S. retail markets: Evidence from big data. *NBER Working Paper*. Available at: `https://www.nber.org/`.

[8] Jensen, R. T., Miller, N. H. (2008). Giffen behavior and subsistence consumption. *American Economic Review*, 98(4), 2008.